Multiple Sclerosis and Related Disorders 93 (2025) 106227

Contents lists available at ScienceDirect

Multiple Sclerosis and Related Disorders

journal homepage: www.elsevier.com/locate/msard

Biomarkers in patients with autoimmune optic neuritis not associated with multiple sclerosis: Demographic, clinical and prognostic features ☆

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ARTICLE INFO

Keywords:
Optic neuritis
Biomarker
Aquaporin 4 antibody
Myelin oligodendrocyte glycoprotein antibody

ABSTRACT

Background: The new optic neuritis (ON) classification leads to a change in how ON patients are grouped. Our aim is to appraise the clinical features and prognoses of patients with autoimmune ON not associated with MS.
Methods: Patients referred to our neuro-ophthalmology laboratory were enrolled to this retrospective study. Patients with ON associated with MS were excluded. The remaining patients were divided into three groups: aqauaporin-4 (AQP4) antibody ON group, myelin oligodendrocyte glycoprotein (MOG) antibody ON group, and seronegative ON group. The patients were examined on admission, one month after acute treatment, and at the third-year follow-up. We compared demographic, clinical, radiologic, laboratory data, and treatment responses among these three groups.
Results: The study included 92 patients. The older age of onset, bilateral simultaneous involvement of the optic nerves, severe vision loss at onset, and need for aggressive treatment were more common in the AQP-ON and the MOG-ON groups than the seronegative ON group ($P = 0.01$, $P = 0.003$, $P = 0.011$, $P = 0.007$, $P < 0.001$, $P < 0.001$, respectively). The presence of optic disc edema was a significant feature of MOG-ON as well as long-length contrast enhancement on MRI ($P = 0.003$, $P = 0.002$). Additional autoimmune antibodies and CNS lesions outside the optic nerve were the features of AQP4-ON patients ($P < 0.001$, $P = 0.015$). Generalized estimating equations analysis revealed that the presence of AQP4 antibody, increased age and recurrence were associated with visual acuity over time ($P = 0.014$, $P = 0.002$, $P = 0.016$, respectively).
Conclusion: The association of serum biomarker status with demographic, clinical features, and visual outcomes indicate the importance of biomarker detection in these patients.

1. Introduction

The term “optic neuritis” (ON) refers to an inflammation of the optic nerve (Kraker and Chen, 2023 Oct). There are numerous etiologies, including autoimmunity, infections and systemic diseases (Kraker and Chen, 2023 Oct). Each cause has different characteristics and requires a distinct clinical approach (Kraker and Chen, 2023 Oct).

A new classification of ON was proposed in 2022 (Petzold et al., Dec). A three-level classification was established, replacing the former perspective of ON as typical and atypical (Petzold et al., 2022 Dec; Abel et al., 2019 Nov-Dec). The new classification dichotomizes ON into two categories at “Level 1″: “autoimmune” and “infectious or systemic”, and defines the subtypes of autoimmune ON at “Level 2″ (Petzold et al., 2022 Dec). The ON etiologies that were previously evaluated under the “atypical optic neuritis” umbrella, such as autoimmune optic neuropathy, infectious optic neuropathy, and ON due to a systemic vasculitis, should now be separated at “Level 1″, according to the new classification (Petzold et al., 2022 Dec).

The majority of ON cases are associated with multiple sclerosis (MS) (Benard-Seguin and Costello, 2023). The new classification has suggested a biomarker and chronology-based approach for patients with autoimmune ON not associated with MS (Petzold et al., 2022 Dec). The identified biomarkers are aquaporin 4 (AQP4) antibody, myelin oligodendrocyte glycoprotein (MOG) antibody, collapsin response mediator protein 5 (CRMP5) antibody and cerebrospinal fluid (CSF) IgG (Petzold et al., 2022 Dec). The biomarkers are helpful in diagnosis, monitoring, and treatment decision (Petzold et al., 2022 Dec).

Several studies have examined the clinical characteristics of ON patients, comparing those with biomarkers to those without (Feng et al., 2021 Jul 16; Ishikawa et al., 2019). However, most studies included

☆ The study was conducted in Ege University Medical School Department of Neurology.

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https://doi.org/10.1016/j.msard.2024.106227
Received 28 October 2024; Received in revised form 8 December 2024; Accepted 9 December 2024
Available online 10 December 2024

patients with MS-ON, and the new classification leads to a change in how patients are grouped (Petzold et al., 2022 Dec; Feng et al., 2021 Jul 16; Ishikawa et al., 2019). Therefore, we believe that there is a need to re-examine patients with autoimmune ON not associated with MS. Our aim is to appraise the clinical features and prognoses of patients with autoimmune ON not associated with MS regarding their biomarker status.

2. Methods

2.1. Participants and data acquisition

This retrospective study was conducted in Ege University Hospital, which is a tertiary referral center in western Turkey. Patients who had been referred to our neuro-ophthalmology laboratory between January 2016 and January 2021 were enrolled. ON Treatment Trial criteria was utilized for diagnosis of ON (Beck et al., 1992). The exclusion and inclusion criteria for the study was designed with the aim of selecting patients with autoimmune ON not associated with MS. The inclusion criteria for our study was as follows: 1) definite diagnosis of ON; 2) older than 18 years of age at the onset; 3) ON symptoms developed within the last 30 days; and 4) being on follow-up for at least three years. Patients with following criteria were excluded from the study: 1) MS diagnosis prior to the ON or during follow-up; 2) not being tested for biomarkers 3) optic neuropathy due to other causes (e.g. vascular, infections, systemic disorders, compression, genetic); 4) incomplete clinical data or follow up; and 5) previously known eye diseases affecting vision. We utilized the McDonald criteria to diagnose patients with MS (Thompson et al., 2018), and also employed the “Infections and systemic disorders associated with optic neuritis” panel in the new classification to exclude ON patients with systemic diseases (Petzold et al., 2022 Dec).

We reviewed the medical records to obtain patients’ data. We noted patients’ age and sex within the demographic data section. We documented the interval from onset to hospital admission, orbital pain at ON presentation, laterality of ON, best corrected visual acuity (BCVA), presence of relative afferent pupillary defect (RAPD), and conducted a fundus examination at each visit, by using a non-mydriatic fundus camera. The patients were examined at least three times: on admission (baseline BCVA), 1 month after acute treatment, and at the third-year follow-up.

All patients underwent orbital magnetic resonance imaging (MRI) with contrast. Longitudinally extensive optic neuritis (LEON) was considered as ON involving more than half the length of the optic nerve (Winter and Chwalisz, 2020). Cranial and spinal MRI findings were noted upon observation of central nervous system (CNS) lesions apart from the optic nerve. Accompanying autoantibodies (anti-nuclear autoantibody [ANA], anti-neutrophil cytoplasmic antibodies [ANCA], anti-Ro, anti-La) were recorded. We also documented ON recurrence, short, and long-term treatment. BCVA was measured in Logarithm of the Minimum Angle of Resolution (LogMAR) scale. According to the latest recommendations in the literature, “counting fingers” was converted to 2.1 LogMAR, “hand movement perception” was converted to 2.4 LogMAR, “light perception” was converted to 2.7 LogMAR and “no light perception” was converted to 3.0 LogMAR (Moussa et al., 2021 Sep). ON was considered bilateral and simultaneous if both eyes were affected within a month. Our primary outcome was BCVA at the third-year follow-up.

ON patients were given 1000 mg methylprednisolone intravenously for five to seven days depending on treatment response. Patients who had had severe vision loss and/or refractory, were administered plasma exchange for five sessions in addition to 1000 mg methylprednisolone. AQP4-ON patients were treated with steroid-sparing immunosuppressants (azathioprine, mycophenolate mofetil) or rituximab following acute treatment. We administered slow oral taper of steroids after high-dose intravenous methylprednisolone for MOG-ON patients, and deployed long-term immunosuppression for patients who had a relapse (s) regardless of the biomarker status.

Third year peripapillary retinal nerve fiber layer (RNFL) thickness and total retinal thickness (measured from the internal limiting membrane [ILM] to the retinal pigment epithelium [RPE] in the central subfield) were scanned. OCT was performed with a spectral domain OCT system (Zeiss Primus 200; Carl Zeiss Meditec, Germany) by an experienced operator. We used 512 × 32 macular cube and 128 × 128 optic disc cube scan for OCT parameter analysis. Images that were of inadequate quality (below 7/10 or showing motion artifacts) were excluded from the analysis.

All patients were tested for serum AQP4 and MOG antibodies using the Euroimmun fixed-cell assay detection kit (Lubeck, Germany). Human MOG and AQP4 isoform M1-transfected HEK293 cells were used in these tests. Patients underwent lumbar punction, and were tested for CSF IgG oligoclonal bands, when necessary. Seronegative ON patients with relapses were tested at each relapse for AQP4 and MOG antibodies. Given that CRMP5-ON is very rare, we investigated the patients for CRMP5 antibody only if suspicion of paraneoplastic ON was present (Yan et al., 2023 Jun 28).

We divided the patients into 3 groups based on their serum biomarker status: AQP4-ON group, MOG-ON group, and seronegative ON group. “Seronegative ON” represents autoimmune ON patients not associated with MS who were negative for both serum AQP4 antibody and serum MOG antibody. We compared demographic and clinical data, radiologic and laboratory findings, and treatment responses among these three groups via pertinent statistical tests.

Local ethical committee of Ege University approved the study (The study code: 24–7T/36). All participants gave informed consent for the study.

2.2. Statistical analysis

We analyzed the data using IBM SPSS version 25.0 (Chicago, IL). Numeric variables, with a normal distribution were defined with means and standard deviations, and those without a normal distribution were defined with median values and interquartile ranges (IQR). Kolmogorov-Smirnov Test was used to test for normality of distribution. In order to compare quantitative data, One-way ANOVA was used for parametric data and the Kruskal-Wallis test for non-parametric data. In order to conduct pairwise multiple comparisons, we used the least significant difference test for parametric data, and Dunn’s test for nonparametric data. In order to compare categorical variables, chi square and Fisher’s exact test were used. Because some patients contributed to the study with two eyes, we took eyes as the intra-subject correlations into our consideration. We created an exchangeable correlation matrix for the intra-subject variable, and formulated a gamma with log link model using GEE. GEE was used with robust variance estimation. All demographic (age, sex), clinical (interval from onset to admission, laterality of ON, presence of orbital pain, optic disc edema, RAPD, and recurrence), radiological (length of contrast enhancement in the optic nerve(s) and CNS lesions other than the optic nerve), laboratory (positive serum ANA/ANCA), and serum biomarker status variables were included in the same GEE model. The dependent variable was LogMAR BCVA. Each patient had three LogMAR BCVA measurements: on admission (baseline BCVA), one month after acute treatment, and at the third-year follow-up. The factors influencing LogMAR BCVA over time were examined using the GEE model with repeated measures. GEE results were presented with regression coefficient (B), standard error (SE), and P values. Values of $P < 0.05$ were considered statistically significant for two-group comparisons. Bonferroni correction was performed for three-group comparisons (0.05/3), and $P < 0.017$ was considered significant.

3. Results

A total of 224 ON patients were evaluated for the study. We excluded

105 MS-ON patients, as well as five patients with toxic optic neuropathy, and four patients with ON due to systemic disorders (one sarcoidosis, one Sjogren's Syndrome, two systemic lupus erythematosus [SLE] patients). Additionally, three infectious optic neuropathy patients and one compressive optic neuropathy patient were excluded. We also excluded 14 ON patients due to incomplete clinical or laboratory data. To sum up, we included 92 patients with a total of 120 eyes in the study. Among the patients included in the study, 19 patients (20.6 %) were positive for serum AQP4 antibodies, 17 (18.5 %) were positive for serum MOG antibodies, and 56 (60.9 %) had a seronegative ON. None of the ON patients tested positive for CRMP5 antibody.

The mean age of AQP4-ON patients was 40.58±10.50 years, MOG-ON patients was 42.41±11.93 years, and seronegative ON patients was 32.45±11.86. The mean age of the seronegative ON patients was significantly lower than that of AQP4-ON and MOG-ON patients (P = 0.010 and P = 0.003, respectively). The female-to-male ratio did not vary significantly among the groups (P = 0.276). However, AQP4-ON group had the highest female-to-male ratio at 3.75.

Bilateral simultaneous ON was present in 9 of 19 (47.3 %) AQP4-ON patients, in 9 of 17 (52.9 %) MOG-ON patients, and 10 of 56 (17.8 %) seronegative ON patients. The occurrence of bilateral simultaneous ON was significantly higher in the AQP4-ON group and the MOG-ON group compared to the seronegative ON group (P = 0.011 and P = 0.007, respectively). Orbital pain at ON presentation was reported in 52.6 % of the AQP4-ON group, 70.5 % of the MOG-ON group, and 55.3 % of the seronegative ON group. The frequency of orbital pain at ON presentation was statistically similar across all groups (P = 0.548).

Optic disc edema was observed in 36.8 % of AQP4 patients, 76.4 % of MOG-ON patients, and 35.7 % of seronegative ON patients. MOG-ON patients exhibited a significantly higher frequency of optic disc edema compared to AQP4-ON patients and seronegative ON patients (P = 0.016 and P = 0.003, respectively). RAPD was least common in the MOG-ON group, occurring in 29.4 % of patients, compared to 63.1 % in the AQP4-ON group and 44.6 % in the seronegative ON group. However, RAPD frequencies in the MOG-ON group did not differ significantly from those in the AQP4-ON and seronegative ON groups (P = 0.042 and P = 0.263, respectively). Relapses occurred in 22.8 % of ON patients overall. Specifically, six of 19 AQP4-ON patients (31.5 %), three of 17 MOG-ON patients (17.6 %), and 12 of 56 seronegative ON patients (21.4 %) had at least one ON relapse. The recurrence rate did not differ significantly among the groups (P = 0.370).

The MOG-ON group showed the highest rate of long length contrast enhancement of the optic nerve. Eight of 17 (47.1 %) MOG-ON patients had long length contrast enhancement in the optic nerve while seven of 56 (12.5 %) seronegative ON patients were found to have it. The difference was statistically significant (P = 0.002). In the AQP4-ON group, five patients (26.3 %) had long length contrast enhancement in the optic nerve.

Three patients (5.3 %) of the seronegative ON group patients displayed a CNS lesion other than the optic nerve on MRI. In contrast, nine of 19 (47.3 %) AQP4-ON patients and four of 17 (23.5 %) MOG-ON patients showed extra-orbital findings on cranial and/or spinal MRI. In particular, six AQP4-ON patients had longitudinally extensive spinal cord lesions (LESCLs), one had a basal ganglia involvement, one had white matter involvement around the fourth ventricle, and one patient had a LESCL and white matter involvement simultaneously. A MOG-ON patient had tumefactive cerebral lesions, one patient had thalamus involvement, one had juxta-cortical lesions and thalamus involvement simultaneously, and the other had a short length myelitis. CNS lesions other than optic nerve was significantly more common in the AQP4-ON group compared to seronegative ON group (P < 0.001).

The AQP4-ON group had the highest rate of autoimmune antibodies at 26.6 %, while the seronegative ON had the lowest at 3.5 % (P = 0.015). Four AQP4-ON patients tested positive for serum ANA, as well as two seronegative ON patients. None of the seronegative ON patients with positive ANA met the criteria for a systemic vasculitis. Additionally, one MOG-ON patient was tested positive for ANCA.

All patients received 1000 mg of intravenous methylprednisolone for five to seven days depending on treatment response. Following this corticosteroid treatment, 16 patients underwent plasma exchange. Additionally, 32 patients were treated with azathioprine, and two patients were given mycophenolate mofetil. Among the AQP4-ON patients, five were treated with rituximab and one with cyclophosphamide. AQP4-ON patients were treated with steroid-sparing immunosuppressants (SSI) more aggressively than MOG-ON patients and seronegative ON patients (P = 0.009 and P < 0.001, respectively). SSI's were also used more frequently in MOG-ON patients compared to seronegative ON patients (P < 0.001). Table 1 provides detailed information about baseline characteristics and treatment of the patients.

Baseline BCVA, BCVA after acute treatment and BCVA at the third-year follow-up were significantly different among the groups. The mean baseline BCVA was 1.0 LogMAR (IQR: 2.275 to 0.045) for AQP4-ON patients, 0.75 LogMAR (IQR: 2.175 to 0.20) for MOG-ON patients, and 0.30 LogMAR (IQR: 0.70 to 0.045) for seronegative ON patients. Seronegative ON patients had better baseline BCVA compared to AQP4-ON patients and MOG-ON patients (P = 0.008 and P = 0.011, respectively). One month after acute treatment, the median BCVA was 0.45 LogMAR (IQR: 1.0 to 0.045) in the AQP4-ON group, 0.10 LogMAR (IQR: 0.30 to 0) in the MOG-ON group, and 0.10 (0.30 to 0) in the seronegative ON group. The median BCVA was significantly higher in the AQP4-ON group compared to the MOG-ON group one month after acute treatment (P = 0.002). At the third-year follow-up, AQP4-ON patients had the worst median BCVA at 0.60 LogMAR (IQR: 2.1 to 0.045) compared to MOG-ON patients with a median BCVA of 0 LogMAR (IQR: 0.045 to 0) and seronegative ON patients with a median BCVA of 0.045 LogMAR (IQR: 0.20 to 0) (P = 0.001 for both comparisons) (Fig. 1).

The AQP4-ON group tended to have lower RNFL and total retinal thickness at the third-year follow-up examination. The mean RNFL thickness was 72.21 μM (SD=11.22), and the mean total retinal thickness was 244.42 μM (SD=14.96) in the AQP4-ON group. However, the difference was not significant across the three groups (P > 0.017). Table 2 shows the LogMAR BCVA and OCT results.

We employed a generalized estimating equations (GEE) model to conduct a longitudinal analysis of BCVA. The analysis revealed that serum antibody biomarker status was significantly associated with LogMAR BCVA over time. MOG-ON patients and seronegative ON patients had significantly lower LogMAR BCVA compared to AQP4-ON patients (B=−0.449, SE=0.171, P = 0.007 and B=−0.386, SE=165, P = 0.014, respectively). A pairwise comparison showed, although MOG-ON patients had lower LogMAR BCVA over time compared to seronegative ON patients, the difference was not statistically significant between MOG-ON patients and seronegative ON patients (B=−0.063, SE=0.112, P = 0.37). Additionally, both increased age and recurrent ON were found to be associated with higher LogMAR BCVA over time (B = 0.015, SE=0.004, P = 0.002 and B = 0.280, SE=−0.123, P = 0.016, respectively). Sex, interval from onset to admission, laterality of ON, presence of orbital pain, optic disc edema, RAPD, long length contrast enhancement in the optic nerve, CNS lesion other than the optic nerve, and positive serum ANA/ANCA had no effect on BCVA over time (P = 0.826, P = 0.252, P = 0.395, P = 0.366, P = 0.946, P = 0.569, P = 0.266, P = 0.062, P = 0.554, respectively). Table 3 provides detailed information regarding covariates and factors influencing LogMAR BCVA longitudinally.

4. Discussion

This study focused on autoimmune ON, particularly the group not associated with MS. Therefore, we excluded almost half of the ON patients because of having MS-ON. MS-ON accounts for 50–80 % of all ON cases, a fact which is compatible with the findings of our study (Benard-Seguin and Costello, 2023). AQP4-ON and MOG-ON constituted 20.6 % and 18.5 % of our study population, respectively. AQP4

Table 1
Baseline characteristics and treatments of the patients.

	AQP4-ON (n = 19, eyes=28)	MOG-ON (n = 17, eyes=26)	Seronegative ON (n = 56, eyes=66)	P_1	P_2	P_3
Age, years (mean±SD)	40.58±10.50	42.41±11.93	32.45±11.86	0.638	**0.010**	**0.003**
Sex, female/male (rate)	15/4 (3.75)	10/7 (1.42)	33/23 (1.43)	0.190	0.116	0.993
Interval of onset to admission, days, median (IQR)	8	5.5 (4 to 9.75)	14 (7.5 to 27)	0.99	**0.012**	**0.006**
Bilateral simultaneous ON, n (%)	9 (47.3)	9 (52.9)	10 (17.8)	0.738	**0.011**	**0.007**
Orbital pain at ON presentation, n (%)	10 (52.6)	12 (70.5)	31 (55.3)	0.269	0.778	0.295
Disc edema, n (%)	7 (36.8)	13 (76.4)	20 (35.7)	**0.016**	0.929	**0.003**
RAPD, n (%)	12 (63.1)	5 (29.4)	25 (44.6)	0.042	0.163	0.263
Recurrent ON, n (%)	6 (31.5)	3 (17.6)	12 (21.4)	0.335	0.370	0.735
Length of optic nerve contrast enhancement, n (%)				0.195	0.155	**0.002**
Short (Less than ½ length of optic nerve)	14 (73.7)	9 (52.9)	49 (87.5)			
Long (More than ½ length of optic nerve)	5 (26.3)	8 (47.1)	7 (12.5)			
CNS lesion other than the optic nerve, n (%)	9 (47.3)*	4 (23.5)*	3 (5.3)	0.137	**<0.001**	0.025
Cortical/Juxta-cortical	0	2 (5.8)	1 (1.7)			
Subcortical	1 (5.2)	2 (5.8)	2 (3.6)			
Brainstem	2 (5.2)	0	0			
Spinal cord	7 (31.5)	1 (5.8)	0			
ANA/ANCA positive, n (%)	4 (26.6)	1 (5.8)	2 (3.5)	0.188	**0.015**	0.674
Plasma exchange, n (%)	7 (36.8)	2 (11.7)	7 (12.5)	0.082	0.181	0.935
Long term treatment, n (%)						
Slow taper of steroids	14 (73.6)	14 (82.3)	26 (46.4)	0.532	0.039	**0.009**
SSI or monoclonal antibody	18 (94.7)	10 (58.8)	7 (12.5)	**0.009**	**<0.001**	**<0.001**

ANA: antinuclear antibody, ANCA: antineutrophil cytoplasmic antibodies, AQP4: aquaporin-4, CNS: central nervous system, IQR: interquartile range, MOG: myelin oligodendrocyte glycoprotein, n: number, ON: optic neuritis, RAPD: Relative Afferent Pupillary Defect, SD: standard deviation, SSI: steroid-sparing immunosuppressant. P_1: P value AQP4-ON compared to MOG-ON, P_2: P value AQP4-ON compared to seronegative ON, P_3: P value MOG-ON compared to seronegative ON.
* : Overlap in some cases.

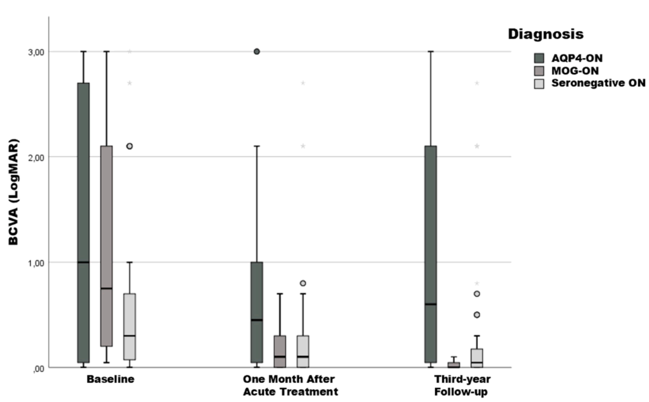

Fig. 1. A clustered boxplot graphic showing patients' visual acuities. BCVA: best corrected visual acuity, LogMAR: Logarithm of the Minimum Angle of Resolution.

antibody was found to show predilection for ethnicity (Petzold et al., 2022 Dec). Although AQP4-ON was responsible for 3 % of ON cases in Caucasians, the rate was reported up to tenfold higher in Asians (Petzold et al., 2022 Dec; Flanagan et al., 2016 May). MOG antibodies cause 5–10 % of adult ON cases worldwide (Petzold et al., 2022 Dec; Ishikawa et al., 2019).

The mean age at onset was the lowest in the seronegative ON group in our study. AQP4 antibodies could typically affect individuals over 40 years of age, and MOG antibodies can affect adults in their 30 s (Chen et al., 2020). The female predominance is most noticeable in the AQP4-ON patients among the ON patients in the literature (Jeyakumar et al., 2024 Aug). Thus, the high female/male ratio in the AQP4-ON group in our study was expected.

Bilateral simultaneous ON and optic disc edema were found most frequently in the MOG-ON group in this study. Bilateral simultaneous ON is a common presentation of MOG-ON, and it was reported up to 84 % in MOG-ON patients (Chen et al., 2020). Optic disc edema is strongly indicative of MOG-ON (Jeyakumar et al., 2024 Aug). Furthermore, MOG-ON can cause papilledema mimicking idiopathic intracranial hypertension (Lotan et al., 2018).

Previous studies found that the highest recurrence frequency was in AQP4-ON patients, and the lowest frequency was in seronegative ON patients (Feng et al., 2021 Jul 16; Lotan et al., 2018). In our study population, the rate of recurrence in AQP4-ON was lower compared to previous studies, and recurrent ON frequency did not differ among the groups (Feng et al., 2021 Jul 16; Lin et al., 2023) We treated AQP4-ON patients aggressively with monoclonal antibodies and SSI's. In the AQP4-ON group, 94.7 % of the patients were given monoclonal antibodies or SSI's. We see this as a viable explanation for the lower recurrence rate in our study.

Previous studies showed that LEON is a distinctive feature for MOG-

Table 2
The LogMAR BCVA and OCT results.

	AQP4-ON	MOG-ON	Seronegative ON	P_1	P_2	P_3
Baseline BCVA, LogMAR, median (IQR)	1.0 (2.275 to 0.045)	0.75 (2.175 to 0.20)	0.30 (0.70 to 0.045)	0.898	**0.008**	**0.011**
BCVA one month after acute treatment, LogMAR, median (IQR)	0.45 (1.0 to 0.045)	0.10 (0.30 to 0)	0.10 (0.30 to 0)	**0.002**	0.018	0.871
BCVA at the third-year follow-up, LogMAR, median (IQR)	0.60 (2.1 to 0.045)	0 (0.045 to 0)	0.045 (0.20 to 0)	**0.001**	**0.001**	0.024
RNFL thickness at the third-year follow-up, μM, mean (SD)	72.21 (11.22)	80.16 (8.13)	83.11 (14.96)	0.354	0.148	0.646
Total retinal thickness at the third-year follow-up, μM, mean (SD)	244.42 (14.96)	248.54 (17.27)	265.46 (24.04)	0.106	0.081	0.738

AQP4: aquaporin-4, BCVA: best-corrected visual acuity, IQR: interquartile range, LogMAR: Logarithm of the Minimum Angle of Resolution, MOG: myelin oligodendrocyte glycoprotein, ON: optic neuritis, RNFL: retinal nerve fiber layer, SD: standard deviation. P_1: P value AQP4-ON compared to MOG-ON, P_2: P value AQP4-ON compared to seronegative ON, P_3: P value MOG-ON compared to seronegative ON.

Table 3
Covariates and factors influencing LogMAR BCVA over time.

	B-Coefficient	Standard Error	*P*
Age, years	0.015	0.004	**0.002**
Sex, male	0.25	−0.121	0.826
Interval of onset to admission	−0.008	0.006	0.252
Bilateral optic neuritis	0.108	−0.116	0.395
Orbital pain	−0.122	−0.018	0.366
Optic disc edema	−0.008	0.111	0.946
RAPD	0.67	−0.117	0.569
Recurrent optic neuritis	0.280	−0.123	**0.016**
More than ½ length of optic nerve contrast enhancement	0.155	0.139	0.266
CNS lesion other than optic nerve	0.291	0.596	0.062
Positive serum ANA/ANCA	0.105	0.445	0.554
Biomarker status			
AQP4	R	R	**0.007**
MOG	−0.449	0.171	**0.014**
Seronegative	−0.386	0.165	

ANA: antinuclear antibody, ANCA: antineutrophil cytoplasmic antibodies, AQP4: aquaporin-4, CNS: central nervous system, IQR: interquartile range, MOG: myelin oligodendrocyte glycoprotein, R: reference, RAPD: relative afferent pupillary defect.

ON and AQP4-ON from MS-ON and other autoimmune ON (Ishikawa et al., 2019; Lin et al., 2023; Ramanathan et al., 2016). In line with the literature, MOG-ON patients in our study population showed the highest rate of LEON. On the other hand, LESCLs are typical for neuromyelitis optica spectrum disorders, and 31.5 % of AQP4-ON patients in our study had LESCL (Jarius et al., 2020). In our study, 26.6 % of AQP4-ON patients had positive serum ANA/ANCA. AQP4-ON was reported to be accompanied by comorbid diseases and autoimmune antibodies more frequently than other autoimmune ONs (Feng et al., 2021 Jul 16; Ishikawa et al., 2019; Nagaishi et al., 2011).

OCT provides quantitative data in ON patients, and that data is correlated with visual acuity (Petzold et al., 2017). Damage to the optic nerve leads to atrophy and consequent thinning in RNFL and macular layers, especially the ganglion cell–inner plexiform layer (GCIPL) (Petzold et al., 2017; Sinnecker et al., 2015). AQP4-ON patients had the worst LogMAR BCVA in our study and tended to have the lowest RNFL thickness on OCT. Besides RNFL thickness, total retinal thickness tended to be lowest in the AQP4-ON group. Thinning in the GCIPL is associated with a decrease in the total retinal thickness (Pilat et al., 2014 Jul). It is regarded as the underlying cause of total retinal thinning in our patients.

We also performed a longitudinal analysis to find out factors associated with BCVA over time and found that a higher age and recurrences were associated with a worse visual outcome in addition to the presence of AQP4 and MOG antibodies. The relationship between increased age and insufficient recovery after ON has been well-documented since ONTT (Beck et al., 1994). This association has also been demonstrated for both MS-ON and other autoimmune patients (Ishikawa et al., 2019; Conway et al., 2019). The AQP4 and MOG antibodies are associated with recurrences, which can affect the outcome of the patient's vision (Jitprapaikulsan et al., 2018).

AQP4-ON patients have the worst outcome among ON patients (Feng et al., 2021 Jul 16; Ishikawa et al., 2019; Lin et al., 2023). In contrast, MOG-ON has a more favorable outcome (Feng et al., 2021 Jul 16; Ishikawa et al., 2019; Lin et al., 2023; Zhao et al., 2018). In this study, AQP4 antibodies were associated with the worst visual outcome in line with the literature. However, the presence of MOG antibodies did not result in a better final visual acuity as expected when compared with the seronegative biomarker status. The explanation can be the exclusion of ON associated with systemic autoimmune disorders including SLE, Sjogren's Syndrome, or sarcoidosis which cause severe vision loss and irrevocable damage from the seronegative group (Lin et al., 2009; Sun et al., 2016 Nov; Kidd et al., 2016 Aug 2). New biomarkers, still waiting to be discovered, could broaden our understanding of ON.

This study has a couple of limitations. Having a retrospective nature, serum biomarkers being tested with commercial fixed-cell detection kits, and measurement of total retinal thickness instead of GCIPL due to our institute facilitations are the main limitations. On the other hand, meticulous patient selection based on new ON classification criteria, who had been on follow-up for at least three years with well documented examination findings is the main strength of the study.

In conclusion, the study identified the characteristics of patients with autoimmune ON not associated with MS. Significant differences between the seropositive and seronegative ON patients were the older age of onset, bilateral simultaneous involvement of the optic nerves, severe vision loss at onset, and need for more aggressive treatment for the seropositive group. The presence of optic disc edema was a significant feature of MOG-ON as well as long-length contrast enhancement on MRI. Additional autoimmune antibodies and CNS lesions outside the optic nerve were the features of AQP4-ON patients. At the end of the third year, though the RNFL or total retinal thickness was not different among groups, AQP4-ON patients had the worst BCVA. Sex, interval from symptom onset to admission, laterality of ON, presence of orbital pain, optic disc edema, RAPD, long-length contrast enhancement in the optic nerve, CNS lesion other than the optic nerve, and positive serum ANA/ANCA had no effect on BCVA over time when the presence of AQP4 and MOG antibodies, increased age and recurrence were the factors associated with poor prognosis. The association of serum biomarker status with demographic and clinical features, as well as visual outcomes, indicate the importance of biomarker detection in these patients. Search for new biomarkers could increase our understanding of ON.

CRediT authorship contribution statement

Hüseyin Nezih Özdemir: Writing – original draft, Visualization, Investigation, Formal analysis. **Mine Topçuoğlu Karakoç:** Writing – original draft, Visualization, Investigation, Data curation. **Figen Gökçay:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Neşe Çelebisoy:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data statement

Our data is available upon a reasonable request.

Acknowledgements

Thanks to Mehmet Mesut Dorukoğlu for helping data acquisition

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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